

CIRCLE ACTIONS ON FUNCTION FIELD BOST–CONNES SYSTEMS: THE COMPACT GROUP ADVANTAGE IS ILLUSORY, AND THE WEIL ZETA FUNCTION IS THERMODYNAMIC

RUQING CHEN

ABSTRACT. In §7 of Paper XVII of this series we suggested that the periodicity of the modular flow over a function field might bring the equivariant Kirchberg classification within reach, compact group actions being better understood than flows. We now check that suggestion and withdraw it.

Takai duality converts a $\mathbb{T}$ -action on $\mathcal{A}$ into a $\mathbb{Z}$ -action on $\mathcal{A} \rtimes \mathbb{T}$, and $\mathbb{Z}$ is discrete and amenable, so the Gabe–Szabó equivariant Kirchberg–Phillips theorem would indeed apply — if $\mathcal{A} \rtimes \mathbb{T}$ were a Kirchberg algebra. For a gauge-type action it is the stabilized core: $\mathcal{O}_n \rtimes \mathbb{T}$ is AF, hence stably finite. The classification theory for Kirchberg algebras therefore does not survive the dualization. Nor does Izumi’s classification of compact group actions apply, that theory requiring the Rokhlin property, whereas gauge actions on Cuntz algebras are approximately representable and not Rokhlin. The advantage we conjectured does not exist.

A second obstruction is arithmetic rather than operator-algebraic. The affine boundary system needs an infinite ring of everywhere-integral elements acting additively, and for a projective curve a function regular at every closed point is constant: $H^0(C, \mathcal{O}_C) = \mathbb{F}_q$, finite. The affine construction degenerates for $S = \mathcal{P}_K$, and one must fix a point at infinity, so the classification cannot be stated for full point sets as proposed.

Third, the equivariant K -theory does not contain the Weil polynomial. The Cuntz relation at P gives $(1 - (qt)^{\deg P})[1] = 0$ in $K_0^{\mathbb{T}}$, so $[1]$ generates the cyclic $R(\mathbb{T})$ -module $\mathbb{Z}[t, t^{-1}]/(1 - (qt)^{d_S})$: this records q^{d_S} and nothing more, the same information already supplied by the point spectrum $\mathrm{Sp}_p(\sigma) = q^{d_S \mathbb{Z}}$ of Proposition 1.1 of Paper XV. The class group contributes $|\mathrm{Pic}^0(C)(\mathbb{F}_q)| = P(1)$, one value of the Weil polynomial rather than the polynomial.

What is true is the conclusion, by another route. The partition function of the Toeplitz system is

$$\sum_{D \in J_S} N(D)^{-\beta} = \sum_{n \geq 0} b_n q^{-\beta n} = Z_C(q^{-\beta}), \quad b_n = \#\{D \geq 0 : \deg D = n\},$$

so the *Weil zeta function is the partition function*. Knowing $\beta \mapsto Z_C(q^{-\beta})$ on $(1, \infty)$ determines the rational function Z_C , hence $P(T)$, the Weil numbers $\{\alpha_j\}$, the genus, every point count $\#C(\mathbb{F}_{q^n}) = q^n + 1 - \sum_j \alpha_j^n$, and $|\mathrm{Jac}(C)(\mathbb{F}_q)| = P(1)$. The complete geometric reconstruction asked for does hold; it is thermodynamic, not K -theoretic — which is the principle of Corollary 2.4 of Paper III appearing once more.

1. THE AFFINE BOUNDARY REQUIRES A POINT AT INFINITY

The framework is that of [5], transported to function fields as in Paper XVII of the series, whose notation we keep. Papers II, III, IX, XIII, XV and XVII, referred to by numeral, are unpublished companions; every fact used from them is restated or verified in place.

Proposition 1.1 (Degeneration for $S = \mathcal{P}_K$). *Let $C/\mathbb{F}_q$ be a smooth projective geometrically irreducible curve and $K = \mathbb{F}_q(C)$. A rational function regular at every closed point of C is constant [3]. Now the affine system $\mathcal{A}^{\mathrm{aff}} = C(\widehat{\mathcal{O}_S}) \rtimes (\mathcal{O}_K \rtimes J_S)$ of Paper II (its §6) requires an infinite additive group of everywhere-integral elements acting densely by translations on the*

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XVIII-equivariant-kirchberg/](#).

profinite completion — by its Remark 6.1 the S -integers themselves have poles at S and do not act, which is why the additive part there is $\mathcal{O}_K$. The function-field counterpart of $\mathcal{O}_K$ at $S = \mathcal{P}_K$ is $H^0(C, \mathcal{O}_C) = \mathbb{F}_q$: the additive group is finite, its closure is a finite set, and the construction degenerates.

Proof. The first statement is that $H^0(C, \mathcal{O}_C) = \mathbb{F}_q$ for C projective and geometrically irreducible. $\square$

Corollary 1.2 (The set-up must be affine). *One must fix a closed point $\infty \in \mathcal{P}_K$ and take $A = \mathcal{O}_{\mathcal{P}_K \setminus \{\infty\}}$, the ring of functions regular away from ∞ , which is a Dedekind domain with $\text{Spec}(A) = \mathcal{P}_K \setminus \{\infty\}$ infinite and dense in $\widehat{\mathcal{O}_S}$ by strong approximation, the omitted place being exactly what the theorem requires; for $C = \mathbb{P}^1$ this is $\mathbb{F}_q[T]$. The classification statement therefore cannot be made for full point sets $S_i = \mathcal{P}_{K_i}$: a choice of ∞ is part of the data, and the partition function of the affine system is $Z_C(T)(1 - T^{\deg \infty})$ rather than $Z_C(T)$.*

2. WITHDRAWAL: THE COMPACT GROUP ADVANTAGE DOES NOT EXIST

Proposition 2.1 (Takai duality lands outside the Kirchberg class). *Let $\alpha : \mathbb{T} \curvearrowright \mathcal{A}$ be the gauge-type action $\alpha_\theta(\mu_{(a,P)}) = e^{i\theta \deg P} \mu_{(a,P)}$ on the affine boundary. By Takai duality [4], α corresponds to the dual action $\hat{\alpha} : \mathbb{Z} \curvearrowright \mathcal{A} \rtimes_\alpha \mathbb{T}$, with $(\mathcal{A} \rtimes \mathbb{T}) \rtimes \mathbb{Z} \cong \mathcal{A} \otimes \mathcal{K}$. The group $\mathbb{Z}$ is discrete and amenable, so the Gabe–Szabó equivariant Kirchberg–Phillips theorem [1] applies to $\mathbb{Z}$ -actions on Kirchberg algebras. However $\mathcal{A} \rtimes_\alpha \mathbb{T}$ is stably finite, and the Kirchberg classification does not apply to the dual system.*

Proof. The action is saturated — there are isometries of every weight in $d_S \mathbb{Z}$ — so $\mathcal{A} \rtimes_\alpha \mathbb{T}$ is Morita equivalent, hence stably isomorphic, to the fixed-point algebra $F = \mathcal{A}^\alpha$. On F the state $\tau = \lambda \circ E$, with λ the Haar probability of $\widehat{\mathcal{O}_S}$ and $E : F \rightarrow C(\widehat{\mathcal{O}_S})$ the canonical expectation, is a faithful trace: traciality reduces to the self-similarity $\lambda(\pi_D U + b) = N(D)^{-1} \lambda(U)$ of Haar measure, the subgroup $\pi_D \widehat{\mathcal{O}_S}$ having index $N(D)$, and this exactly cancels the $N(D)$ cosets appearing in $E(vxv^*)$ for a weight-deg D isometry v . A unital C^* -algebra with a faithful tracial state is stably finite, and stable finiteness passes to the stabilization. The model case is classical: already $\mathcal{O}_n \rtimes_\gamma \mathbb{T} \cong \mathcal{K} \otimes (\text{UHF core})$ is AF. $\square$

Proposition 2.2 (The action is not Rokhlin). *Izumi’s classification of finite and compact group actions on Kirchberg algebras requires the Rokhlin property [2]. A Rokhlin action of $\mathbb{T}$ on a unital $\mathcal{A}$ forces $\mathcal{A}$ to absorb the fixed-point algebra in a strong sense; the restrictions of gauge actions on Cuntz and Cuntz–Pimsner algebras to finite cyclic subgroups are instead approximately representable, the notion dual to Rokhlin [2], and the circle action itself does not have the Rokhlin property. So that route is unavailable as well.*

Remark 2.3 (Withdrawal of a suggestion). We suggested there that periodicity of σ “may bring Remark 5.1 of Paper XIII within reach”, on the ground that the equivariant classification for compact groups is better developed than for $\mathbb{R}$. That is true as a general statement about the literature and false as an inference here, for the two reasons above: dualizing a circle action moves the problem to a stably finite algebra, and the specific action fails the Rokhlin hypothesis under which the compact group theory is available. We withdraw the suggestion. The caution of Remark 5.1 of Paper XIII, that flows on Kirchberg algebras are not classified, stands and is not circumvented by compactness.

3. WHAT THE EQUIVARIANT K -THEORY RECORDS

Proposition 3.1 (The cyclic module). *In $K_0^{\mathbb{T}}(\partial \mathcal{A}^{\text{aff}})$, the Cuntz relation at P — the sum over the $NP = q^{\deg P}$ additive cosets of the range projections, each of weight $t^{\deg P}$ — gives*

$$(1 - (qt)^{\deg P}) [1] = 0 \quad (P \in S).$$

Since $x = qt$ becomes a unit modulo each relation $1 - x^{\deg P}$, the exponents combine by Bézout: $\gcd_i(1 - x^{n_i}) = 1 - x^{\gcd_i n_i}$ as ideals of $\mathbb{Z}[t, t^{-1}]$, and the class $[1]$ generates the cyclic $R(\mathbb{T}) = \mathbb{Z}[t, t^{-1}]$ -module

$$R(\mathbb{T}) \cdot [1] \cong \mathbb{Z}[t, t^{-1}] / (1 - (qt)^{d_S}), \quad d_S = \gcd_{P \in S} \deg P.$$

Corollary 3.2 (No genus). *The module of Proposition 3.1 records the single quantity q^{d_S} , which is exactly the content of the point spectrum $\text{Sp}_p(\sigma) = q^{d_S \mathbb{Z}}$ already available from $(\mathcal{A}, \sigma)$ by Proposition 1.1 of Paper XV. It contains no information about the genus, the Weil numbers, or the point counts.*

Remark 3.3 (What K -theory does see). By Theorem 3.3 of Paper II the class of the unit in K_0 of the boundary records the narrow class number; the counterpart of that quantity here is $|\text{Pic}^0(C)(\mathbb{F}_q)| = |\text{Jac}(C)(\mathbb{F}_q)|$, of order $P(1)$. So K -theory does see one value of the Weil polynomial, at $T = 1$. It does not see the polynomial: the $2g$ numbers α_j are not determined by $P(1)$, and no mechanism in the six-term sequence produces the higher point counts.

4. THE WEIL ZETA FUNCTION IS THE PARTITION FUNCTION

Theorem 4.1 (Z_C as a thermodynamic quantity). *Let $b_n = \#\{D \in \text{Div}_+(C) : \deg D = n\}$. Then, by the definition of the Weil zeta function,*

$$Z_C(T) = \sum_{n \geq 0} b_n T^n = \frac{P(T)}{(1 - T)(1 - qT)}, \quad P(T) = \prod_{j=1}^{2g} (1 - \alpha_j T),$$

and the partition function of the Toeplitz system $(\mathcal{A}_{C, \mathcal{P}_K}, \sigma)$ is

$$(4.1) \quad \sum_{D \in J_{\mathcal{P}_K}} N(D)^{-\beta} = \sum_{n \geq 0} b_n q^{-\beta n} = Z_C(q^{-\beta}), \quad \beta > 1.$$

Equivalently, the $\mathbb{T}$ -action on the Fock space $\ell^2(J_S)$ has weight- n multiplicity b_n , and Z_C is its graded dimension.

Proof. $N(D) = q^{\deg D}$ and $J_{\mathcal{P}_K} = \text{Div}_+(C)$; group the sum by degree. Convergence for $\beta > 1$ is Corollary 2.2 of Paper XVII. $\square$

Theorem 4.2 (Complete geometric reconstruction). *The function $\beta \mapsto Z_C(q^{-\beta})$ on $(1, \infty)$ determines the rational function Z_C , hence:*

- (1) *the Weil polynomial $P(T)$ and the Weil numbers $\{\alpha_1, \dots, \alpha_{2g}\}$;*
- (2) *the genus $g = \frac{1}{2} \deg P$;*
- (3) *every point count $\#C(\mathbb{F}_{q^n}) = q^n + 1 - \sum_{j=1}^{2g} \alpha_j^n$;*
- (4) *the order $|\text{Jac}(C)(\mathbb{F}_q)| = P(1)$ of the group of rational points of the Jacobian.*

Consequently the equivariant C^ -dynamical system $(\mathcal{A}_{C, \mathcal{P}_K}, \mathbb{T}, \sigma)$ is a complete invariant of the Weil zeta function of C .*

Proof. A rational function is determined by its values on an interval; q is recovered from the point spectrum $\text{Sp}_p(\sigma) = q^{d_S \mathbb{Z}}$ (Proposition 1.1 of Paper XV, computed for these systems in Remark 4.3 of Paper XVII), with $d_S = 1$ for $S = \mathcal{P}_K$ by F. K. Schmidt (Proposition 1.1 of Paper XVII), and then $P(T) = Z_C(T)(1 - T)(1 - qT)$. The remaining items are the Weil conjectures for curves. $\square$

Example 4.3 (Numerical illustration). For genus one, $P(T) = 1 - aT + qT^2$:

(q, a)	$b_0, \dots, b_4$	$\#C(\mathbb{F}_{q^n}), n \leq 4$	$ \text{Jac}(\mathbb{F}_q) $
$(4, 1)$	1, 4, 20, 84, 340	4, 24, 76, 240	4
$(5, -2)$	1, 8, 48, 248, 1248	8, 32, 104, 640	8
$(9, 4)$	1, 6, 60, 546, 4920	6, 84, 774, 6720	6

The b_n are the multiplicities in the partition function (4.1); the point counts and the Jacobian order follow from them.

Remark 4.4 (The principle, again). Theorem 4.2 delivers the conclusion asked for while Corollary 3.2 shows the proposed route cannot. This is the pattern established in Corollary 2.4 of Paper III: the arithmetic of these systems lives in the equilibrium data — the measure class, the KMS simplex, the partition function — and K -theoretic invariants, which discard the Radon–Nikodym derivative, are blind to it. Over a function field the equilibrium datum happens to be the Weil zeta function itself, which is why the reconstruction is so complete; but the mechanism is thermodynamic, and asking $KK_{\mathbb{T}}$ to supply it repeats an error the programme has already diagnosed three times, in Remark 4.5 of Paper II, Proposition 3.1 of Paper III and Theorem 2.1 of Paper IX.

5. ASSESSMENT

Proposal	Status	Reference
$\partial\mathcal{A}^{\text{aff}}$ defined for $S = \mathcal{P}_K$	no: $\mathcal{O}_{\mathcal{P}_K} = \mathbb{F}_q$	Prop. 1.1
a point at infinity must be fixed	yes	Cor. 1.2
$\mathbb{T}$ -actions better behaved than flows	not here	Prop. 2.1
Gabe–Szabó applies after Takai	no: dual algebra stably finite	Prop. 2.1
Izumi’s compact group theory applies	no: action not Rokhlin	Prop. 2.2
§7 of Paper XVII suggestion	withdrawn	Rem. 2.3
$K_0^{\mathbb{T}}$ encodes $P(T)$	no: only q^{ds}	Prop. 3.1, Cor. 3.2
K -theory sees $ \text{Jac}(\mathbb{F}_q) = P(1)$	yes, one value only	Rem. 3.3
Z_C is the partition function	yes	Thm. 4.1
complete geometric reconstruction	yes, thermodynamically	Thm. 4.2

Two of the three obstructions are specific and one is a retraction of our own suggestion. The retraction is worth stating plainly: the observation that compact group actions are better classified than flows is correct about the literature and inapplicable here, because the standard reduction — Takai duality to a $\mathbb{Z}$ -action — lands on the stably finite core, and the compact group theorems that avoid that reduction require a Rokhlin property that gauge actions do not have. Periodicity of the modular flow is a real structural feature of the function field case, but it does not buy a classification theorem.

The positive statement is stronger than what was asked for the boundary, and it needs no classification theory at all. Over a function field the partition function of the Toeplitz system *is* the Weil zeta function, evaluated at $T = q^{-\beta}$; and since Z_C is rational, its values on any interval determine everything the Weil conjectures encode. The system therefore recovers g , all $\#C(\mathbb{F}_{q^n})$, the Weil numbers and $|\text{Jac}(C)(\mathbb{F}_q)|$ — from thermodynamics.

Three questions. Does the affine boundary with a fixed ∞ recover $\deg \infty$, so that Z_C can be reconstructed from the affine partition function $Z_C(T)(1 - T^{\deg \infty})$? Second, is there any equivariant invariant, short of the full KMS structure, that sees the b_n — the graded dimension of the Fock module is the natural candidate, but we do not know whether it is an invariant of the abstract dynamical system rather than of the chosen representation. Third, and this is the one

we would pursue: two curves with the same zeta function but non-isomorphic Jacobians exist; by Theorem 4.2 their systems have the same partition function, so does anything in $(\mathcal{A}, \mathbb{T}, \sigma)$ distinguish them, or is zeta-equivalence exactly the equivalence realized here?

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GUT GEOSERVICE INC., MONTREAL, CANADA
Email address: ruqing@hotmail.com